Response to Reviewers' Comments

Paper ID 2147 • ICMLDE 5.0 • Procedia Computer Science

Original title: *Edge-Deployable Skin Lesion Classification: A Weighted Ensemble of MobileNetV2 and DINOv2*

Revised title: *Edge-Deployable Skin Lesion Classification with Quantized CNN and Transformer Models*

Respected Scientific Committee and Reviewers,

We sincerely thank the reviewers for their careful reading and constructive comments, which have substantially strengthened the paper. In addressing them, we carried out a detailed audit of the data split and of the model-fusion evaluation, which sharpened the focus of the contribution. The revised paper presents MobileNetV2 and DINOv2 as independent branches, analyses their probability-level fusion with a paired statistical test, and adds on-device latency, memory, and thermal measurements on a Raspberry Pi 4 for the FP32 and quantized models. All accuracies are reported transparently as validation results, together with a lesion-level overlap analysis.

Summary of Major Changes

1. **Title.** Revised from "Edge-Deployable Skin Lesion Classification: A Weighted Ensemble of MobileNetV2 and DINOv2" to "Edge-Deployable Skin Lesion Classification with Quantized CNN and Transformer Models".

   Justification: The new title reflects the central contribution of the revision, quantized models evaluated for edge deployment (Reviewer 1, Comments 2 and 3; Reviewer 3, Comment 1).

2. **Block diagram (Fig. 1 (page 3)).** Redrawn to show data preparation, the two MobileNetV2 and DINOv2 branches with their baseline FP32 evaluation and optimization experiments, Grad-CAM explainability, the three output classes, and the Raspberry Pi 4 deployment with its measured latency.

   Justification: The diagram now presents the complete revised workflow in one view, including the optimization, explainability, and on-device deployment stages requested by the reviewers (Reviewer 1, Comment 3; Reviewer 3, Comments 1 and 2).

3. **Methodology.** The two models are presented as independent branches, and probability-level fusion is analysed in a dedicated section with a paired McNemar test (Section 3.6 (page 4) and 4.4).

   Justification: Independent branches give a clear, deployable model, and the dedicated fusion analysis quantifies the benefit of combining the two architectures with a statistical test (Reviewer 1, Comments 2, 4, and 6; Reviewer 2, Comment 5).

4. **Edge deployment.** New Section 4.6 (page 7) and Table 4 (page 7) report Raspberry Pi 4 latency, throughput, peak memory, and sustained thermal behaviour for the FP32 and quantized models.

   Justification: These on-device measurements provide direct evidence for the edge-deployable claim and show the effect of quantization on a real edge device (Reviewer 1, Comment 3; Reviewer 3, Comment 1).

5. **Data analysis.** Lesion-level overlap analysis (Section 3.2 (page 4)), justification of the three classes, and class-balance statistics (Section 3.1 (page 3)).

Justification: The dataset composition and the data split are now described transparently, so that the scope of the results is clear to the reader (Reviewer 1, Comment 1; Reviewer 2, Comments 1, 3, and 4).

6. **Graphs and figures.** All result graphs were regenerated from the experiment logs in a consistent publication style (bold labels, inward ticks, boxed legends):

   - Training curves: the two separate MobileNetV2 and DINOv2 accuracy/loss graphs were combined into one four-panel figure (Fig. 3 (page 6)).
   - Confusion matrix: replaced by the MobileNetV2 FP32 confusion matrix, with a matching per-class table (Fig. 4 (page 8), Table 5 (page 8)).
   - Grad-CAM: new figure with heatmaps for one correct case per class and a Melanoma-predicted-as-Nevus case (Fig. 5 (page 9)).
   - Block diagram: redrawn as described in item 2 (Fig. 1 (page 3)).
   - The earlier sample-image figure was merged into Figs. 1 and 5, which show the same lesions.

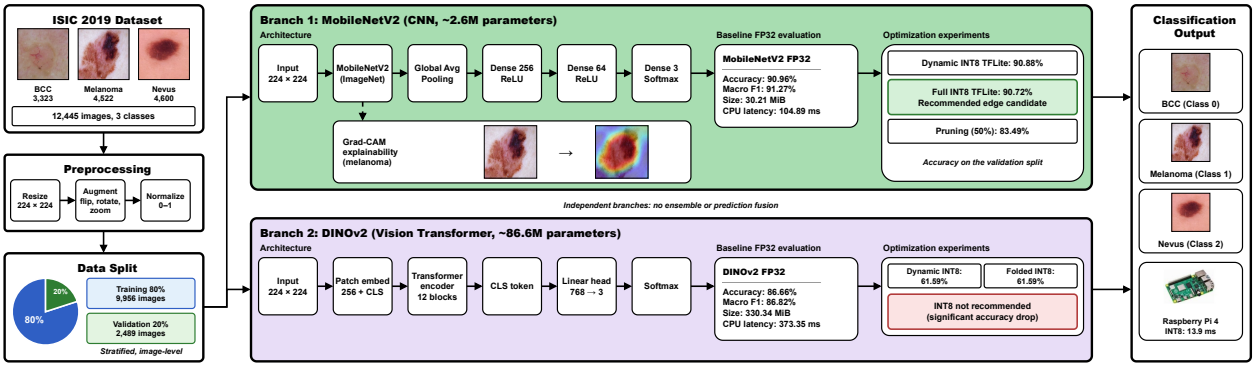
   Justification: The regenerated graphs report the deployed MobileNetV2 directly, add per-class and Melanoma-specific evidence and visual explanations, and present the training behaviour of both models side by side in less space, which keeps the manuscript within the ten-page limit (Reviewer 2, Comment 2; Reviewer 3, Comments 2 and 3).

7. **Text.** Abstract, introduction, contributions, discussion, and conclusion rewritten to reflect these changes.

   Justification: The narrative is now consistent with the revised results, discusses the fusion gain, and states the study scope and future work explicitly (Reviewer 1, Comment 6; Reviewer 2, Comment 6; Reviewer 3, Comment 6).

8. **References.** Seven references added, four previously uncited references now cited, and the list numbered in order of citation.

   Justification: The quantization, pruning, explainability, and transformer-quantization methods used in the study are now supported by the established literature (Reviewer 3, Comment 5).



Revised block diagram (Figure 1, page 3, of the revised manuscript).

Point-by-Point Responses

All section, figure, table, and page numbers in this letter refer to the revised manuscript, which follows this letter.

Reviewer 1

Reviewer 1, Comment 1: *The abstract reports validation accuracy; independent test results should be reported.*

Response: We thank the reviewer for this important observation. The abstract and all results now state clearly that they are validation results, together with a lesion-level overlap analysis (54.04% of validation images share a lesion with the training set). To give the most representative estimate, we also report accuracy on the 1,144 validation images whose lesions are distinct from training (MobileNetV2 86.80%, DINOv2 84.00%). A lesion-grouped split has been prepared for the held-out test evaluation planned as future work.

Accuracy on the full and the non-overlapping validation images

Model	All 2,489 images	1,144 non-overlapping images
MobileNetV2 FP32	90.96%	86.80%
DINOv2 FP32	86.66%	84.00%
Equal-weight fusion	91.36%	87.15%

Changes in the Manuscript (Abstract, page 1; Section 3.2, page 4 of the revised manuscript): "... *All results are validation results, because lesion-level auditing identified 54.04% overlap between validation and training images. ...*"

Reviewer 1, Comment 2: *A weighted ensemble of MobileNetV2 and DINOv2 is a modest contribution.*

Response: We thank the reviewer for this valuable comment. In response, the contribution has been redefined around edge deployment: the revised paper presents (i) a controlled comparison of FP32, dynamic-range INT8, full-integer INT8, and 50% pruned MobileNetV2; (ii) on-device latency, memory, and thermal measurements on a Raspberry Pi 4; and (iii) an analysis of INT8 quantization for DINOv2, including the activation outliers in Blocks 8–11. The probability-level fusion is retained as a supporting analysis, and the title now reflects quantization as the central subject.

Changes in the Manuscript (Title; Section 1, page 2 of the revised manuscript): "... *Integration of the MobileNetV2 inference pipeline into a Streamlit-based application, and on-device benchmarking of the FP32 and quantized MobileNetV2 variants on a Raspberry Pi 4, including latency, memory, and sustained thermal behaviour. ...*"

Reviewer 1, Comment 3: *The edge-deployable claim needs latency and memory measurements on edge devices.*

Response: We thank the reviewer for this suggestion. All MobileNetV2 variants were benchmarked on a Raspberry Pi 4 Model B (4 GB, Cortex-A72, CPU only). Full INT8 TFLite ran at 13.92 ms per image (71.8 FPS) with 51.7 MiB peak memory, compared with 597.59 ms and 636.9 MiB for the FP32 Keras model; an unquantized FP32 TFLite baseline (36.63 ms) separates the runtime effect from the quantization effect. Sustained 5-minute runs reached at most 78.4 °C with no throttling, and the Full INT8 latency remained constant (14.11 to 14.16 ms).

Raspberry Pi 4 Model B results (batch one, four threads)

Variant	Disk	Latency (ms)	FPS	Peak RSS	Max temp.
FP32 Keras	30.21 MiB	597.59 ± 9.86	1.67	636.9 MiB	55.0 °C
FP32 TFLite	9.79 MiB	36.63 ± 0.34	27.30	69.5 MiB	74.5 °C
Dynamic-range INT8	2.74 MiB	41.72 ± 3.42	23.97	59.2 MiB	72.0 °C
Full INT8	**2.94 MiB**	**13.92 ± 0.20**	**71.83**	**51.7 MiB**	**78.4 °C**

Changes in the Manuscript (Section 4.6, Table 4, page 7 of the revised manuscript): "*. . . Full INT8 TFLite ran at 13.92 ms per image (71.8 FPS), 42.9× faster than FP32 Keras and 2.6× faster than unquantized FP32 TFLite, while using the least memory. . . .*"

Reviewer 1, Comment 4: *Ensemble weights tuned on validation data must be tested on a separate test set.*

Response: We thank the reviewer for this point. Weighted fusion is reported as a dedicated analysis in Section 4.4 (page 6), which compares both evaluated weightings with MobileNetV2 on the same validation predictions, while the deployed pipeline uses the Full INT8 MobileNetV2. Evaluation of the fusion weights on a separate lesion-disjoint test set is planned as future work.

Changes in the Manuscript (Section 4.4, Table 2, page 6 of the revised manuscript): "*. . . Table 2 sets MobileNetV2 against the two fusion weightings we tried. . . .*"

Reviewer 1, Comment 5: *Comparison with published ISIC 2019 results is needed.*

Response: We thank the reviewer for this suggestion. Published ISIC 2019 studies address the full eight-class task, mostly with balanced accuracy and with their own data splits, while this study focuses on a three-class subset. To keep every comparison like-for-like, the revised manuscript states the task scope explicitly and presents a controlled comparison of model and compression variants under one common protocol, in which every number is directly comparable.

Changes in the Manuscript (Abstract, page 1 of the revised manuscript): "*. . . The study focuses on a reproducible comparison of full-precision, quantized, and pruned models, together with a Grad-CAM explainability assessment. . . .*"

Reviewer 1, Comment 6: *The conclusion should discuss the small 1.24-point gain.*

Response: We thank the reviewer for this comment. Based on the verified predictions, the equal-weight fusion improves on MobileNetV2 by 0.40 points (10 additional correct images; McNemar $p = 0.174$), and the 0.4/0.6 configuration reaches 90.76%. The Discussion and Conclusion now discuss this gain and explain why the Full INT8 MobileNetV2, which combines high accuracy with a 2.94 MiB model and 13.92 ms on-device latency, is selected as the deployment model.

Fusion analysis on the 2,489 validation images

Configuration	Accuracy	Macro F1	Melanoma recall
MobileNetV2 FP32	90.96%	91.27%	87.06%
DINOv2 FP32	86.66%	86.82%	79.09%
Fusion ($w_M = 0.5$, $w_D = 0.5$)	91.36%	91.67%	87.39%
Fusion ($w_M = 0.4$, $w_D = 0.6$)	90.76%	91.01%	85.18%

Changes in the Manuscript (Section 5, page 8 of the revised manuscript): "... *On a Raspberry Pi 4 it ran at 71.8 FPS, which confirms Full INT8 MobileNetV2 as a practical edge model.* ..."

Reviewer 2

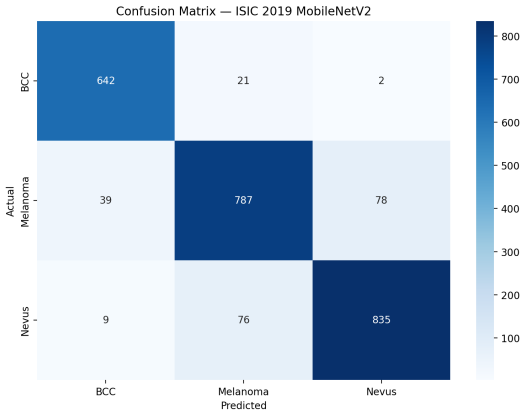
Reviewer 2, Comment 1: *Only three classes are used; the selection should be justified.*

Response: We thank the reviewer for this suggestion. The three classes were chosen because they cover a malignant melanocytic lesion (Melanoma), its most common benign look-alike (Nevus), and the most common non-melanocytic skin cancer (BCC). This isolates the clinically most consequential Melanoma–Nevus confusion and keeps the class distribution comparatively balanced. The limitations of this choice are stated in the Discussion.

Changes in the Manuscript (Section 3.1, page 3 of the revised manuscript): "... *They cover a malignant melanocytic lesion, its most common benign look-alike, and the most common non-melanocytic skin cancer, which isolates the clinically most consequential Melanoma–Nevus confusion.* ..."

Reviewer 2, Comment 2: *Per-class recall, especially for melanoma, should be reported.*

Response: We thank the reviewer for this suggestion. Table 5 (page 8) reports per-class precision, recall, and F1 for MobileNetV2 (Melanoma recall 87.06%), Table 3 (page 7) reports Melanoma recall for every model variant, and the confusion matrix in Fig. 4 (page 8) shows that the main error is Melanoma predicted as Nevus (78 images).



Confusion matrix of MobileNetV2 FP32 (Figure 4, page 8, of the revised manuscript).

Class-level performance of MobileNetV2 FP32

Class	Support	Precision	Recall	F1-score
BCC	665	93.04%	96.54%	94.76%
Melanoma	904	89.03%	87.06%	88.03%
Nevus	920	91.26%	90.76%	91.01%
Macro average	2,489	91.11%	91.45%	91.27%

Changes in the Manuscript (Section 4.7, Figure 4, Table 5, page 8 of the revised manuscript): "*. . . BCC achieved the highest recall, while Melanoma achieved the lowest. . . .*"

Reviewer 2, Comment 3: *Class balance should be described.*

Response: We thank the reviewer for this suggestion. The dataset contains 3,323 BCC, 4,522 Melanoma, and 4,600 Nevus images; the stratified split gives 2,658 / 3,618 / 3,680 training and 665 / 904 / 920 validation images. The classes are therefore close to balanced (largest-to-smallest ratio 1.38), and the stratified split preserves these proportions in both partitions.

Class distribution

Class	Total	Training (80%)	Validation (20%)
BCC	3,323	2,658	665
Melanoma	4,522	3,618	904
Nevus	4,600	3,680	920
Total	**12,445**	**9,956**	**2,489**

Changes in the Manuscript (Section 3.1, page 3 of the revised manuscript): "*. . . With 3,323 BCC, 4,522 Melanoma, and 4,600 Nevus images, the largest class is only 1.38 times the smallest, so the subset is close to balanced. . . .*"

Reviewer 2, Comment 4: *Lesion-level splitting should be confirmed.*

Response: We thank the reviewer for this important point. A lesion-ID audit was carried out and is now reported in Section 3.2 (page 4): the split is image-level, and 54.04% of validation images belong to lesions present in training (BCC 85.11%, Melanoma 79.31%, Nevus 6.74%). The manuscript reports this transparently, gives results on the non-overlapping subset, and a lesion-grouped split has been prepared for retraining in future work.

Share of validation images whose lesion also appears in training

Class	Validation images with lesion overlap
BCC	85.11%
Melanoma	79.31%
Nevus	6.74%
All classes	**54.04%**

Changes in the Manuscript (Section 3.2, page 4 of the revised manuscript): "... *Of the validation images, 1,345 (54.04%) belonged to lesions also represented in the training partition.* ..."

Reviewer 2, Comment 5: *Statistical significance should be tested.*

Response: We thank the reviewer for this suggestion. An exact two-sided McNemar test now compares MobileNetV2 with the equal-weight fusion on the same 2,489 images: the fusion corrected 27 MobileNetV2 errors and changed 17 correct predictions ($p = 0.174$), and it raised recall in every class. The manuscript reports this test together with the per-class gains, and confidence intervals on a lesion-disjoint test set are planned as future work.

Paired outcomes of MobileNetV2 and the equal-weight fusion (McNemar)

	Fusion correct	Fusion wrong
MobileNetV2 correct	2,247	17
MobileNetV2 wrong	27	198

Exact two-sided McNemar test on the discordant pairs (27 vs. 17): $p = 0.174$.

Changes in the Manuscript (Section 4.4, page 6 of the revised manuscript): "... *Equal-weight fusion corrected 27 MobileNetV2 errors and introduced 17 new ones, a net gain of 10 images, and raised recall in every class (BCC* +0.30*, Melanoma* +0.33*, Nevus* +0.54 *percentage points).* ..."

Reviewer 2, Comment 6: *Limitations of using a three-class subset of ISIC 2019 should be stated.*

Response: We thank the reviewer for this suggestion. The Discussion now describes the scope of the study clearly: three of the eight ISIC 2019 categories, 1,886 images without lesion IDs, and external multi-centre validation as a direction for future work. The results are accordingly presented as a comparative study of model and compression variants.

Changes in the Manuscript (Section 7, page 9 of the revised manuscript): "... *This work presented a comparative study of MobileNetV2 and DINOv2 for three-class skin lesion classification.* ..."

Reviewer 3

Reviewer 3, Comment 1: *DINOv2 deployment on edge devices should be discussed.*

Response: We thank the reviewer for this suggestion. DINOv2 FP32 requires 330.34 MiB and 235–373 ms per image on the development CPU. Native dynamic INT8 reduces it to 87.38 MiB and 173.54 ms, with accuracy of 61.59%. Layer-wise diagnostics trace this to activation outliers in Blocks 8–11, a known challenge for transformer quantization, and quantization-aware training or distillation to a smaller ViT is identified as the route to edge deployment.

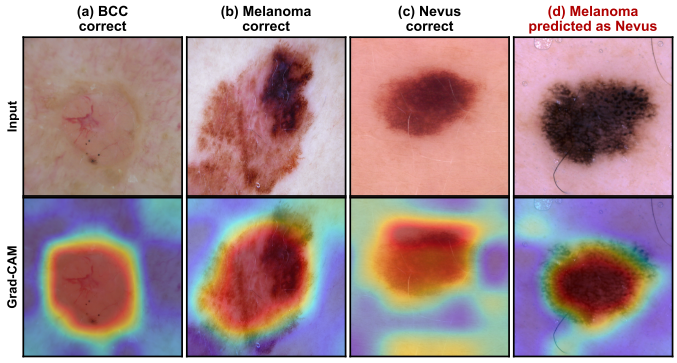
DINOv2 deployment characteristics

Variant	Disk	Latency (dev. CPU)	Accuracy	Melanoma recall
DINOv2 FP32	330.34 MiB	235–373 ms	86.66%	79.09%
DINOv2 dynamic INT8	87.38 MiB	173.54 ms	61.59%	49.67%
MobileNetV2 Full INT8 (reference)	2.94 MiB	4.51 ms	90.72%	86.73%

Changes in the Manuscript (Section 4.5, page 7 of the revised manuscript): "... *Layer-wise diagnostics identified extreme activation outliers in transformer Blocks 8–11, a known obstacle to post-training transformer quantization, with magnitudes up to approximately 494.72 compared with upper-percentile values near 1.3–2.7 in the corresponding diagnostic summaries. ...*"

Reviewer 3, Comment 2: *Grad-CAM visualisations would help.*

Response: We thank the reviewer for this suggestion. A new Grad-CAM section and figure show MobileNetV2 heatmaps for one correctly classified image of each class and for a Melanoma image predicted as Nevus. In the correct cases the activation is concentrated on the lesion; in the misclassified case the model also attends to the lesion, indicating that the error arises from interpreting the lesion rather than from attention to irrelevant regions.

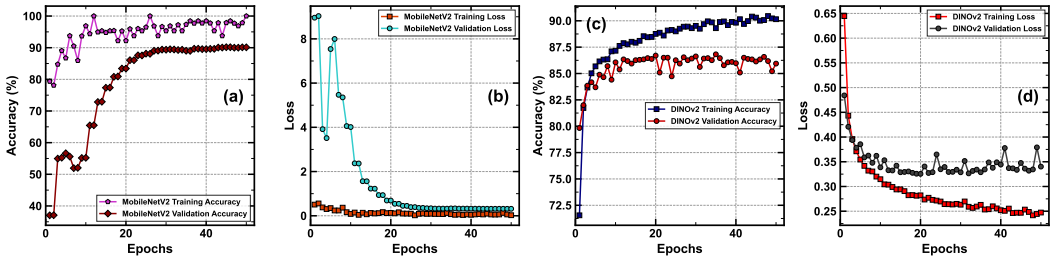


Grad-CAM heatmaps of MobileNetV2 (Figure 5, page 9, of the revised manuscript): (a)–(c) correct BCC, Melanoma, and Nevus; (d) Melanoma predicted as Nevus.

Changes in the Manuscript (Section 6, Figure 5, page 9 of the revised manuscript): "... *In the failure case, the model also attends to the lesion but still predicts Nevus, which indicates that the error arises from how the lesion is interpreted rather than from attention to irrelevant regions. ...*"

Reviewer 3, Comment 3: *Training settings should be reported.*

Response: We thank the reviewer for this suggestion. The training settings are reported in the methodology: input size (224×224), normalization for each model, the augmentations used (rotation, zoom, translation), ImageNet and self-supervised pretraining, the classification heads, and 50 training epochs for both models, with the epoch-wise curves in Fig. 3. Owing to the ten-page limit, they are given in the text rather than in a separate table.



Training and validation curves of MobileNetV2 (a, b) and DINOv2 (c, d) over 50 epochs (Figure 3, page 6, of the revised manuscript).

Changes in the Manuscript (Section 3.3, page 4 of the revised manuscript): "... *MobileNetV2 was trained with rotation, zoom, and translation augmentation. ...*"

Reviewer 3, Comment 4: *Fairness across skin tones should be discussed.*

Response: We thank the reviewer for raising this important point. ISIC 2019 does not include Fitzpatrick skin-type labels, and a reliable subgroup analysis requires a dataset with annotated skin types. The manuscript discusses this explicitly, and a skin-tone fairness evaluation on such a dataset is included in the planned future work.

Changes in the Manuscript (Section 7, page 9 of the revised manuscript): "... *Further work should include paired statistical tests, confidence intervals, power measurements and testing on additional edge devices, attention rollout for DINOv2, skin-tone fairness analysis, calibration, and comparison with external datasets. ...*"

Reviewer 3, Comment 5: *Recent edge-oriented skin lesion classifiers should be cited.*

Response: We thank the reviewer for this suggestion. The revised manuscript now cites the efficient-model and compression literature on which the edge results build: MobileNet and MobileNetV2, integer-only quantization, post-training quantization, magnitude pruning, and transformer quantization. Because the edge claim rests on direct on-device measurements, the work is positioned against these methods rather than against systems evaluated on different devices, datasets, and class sets.

Changes in the Manuscript (Section 4.5, page 7 of the revised manuscript): "... *Layer-wise diagnostics identified extreme activation outliers in transformer Blocks 8–11, a known obstacle to post-training transformer quantization, with magnitudes up to approximately 494.72 compared with upper-percentile values near 1.3–2.7 in the corresponding diagnostic summaries. ...*"

Reviewer 3, Comment 6: *Future work could include all ISIC classes.*

Response: We thank the reviewer for this suggestion. Future work now includes extending the study to all eight ISIC 2019 categories, together with lesion-disjoint retraining, quantization-aware training, and ViT-S distillation.

Changes in the Manuscript (Section 7, page 9 of the revised manuscript): "... *Extending the study to all eight ISIC 2019 categories, quantization-aware training, ViT-S distillation, out-of-distribution rejection, and imbalance-aware losses may improve the reliability of a future deployment-oriented system.* ..."

Editorial requirements

- **Figures:** all figures were produced by the authors for this manuscript from their own experiments.
- **References:** seven references were added (quantization, pruning, Grad-CAM, TensorFlow, transformer activation outliers), four previously uncited references are now cited, and the list is numbered in order of citation.

Sincerely,
Darapureddi Chetan Kumar and Jinka Venkata Aravind

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Procedia Computer Science 00 (2026) 000–000

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5th International Conference on Machine Learning and Data Engineering

Edge-Deployable Skin Lesion Classification with Quantized CNN and Transformer Models

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Abstract

Automated classification of skin lesions using dermoscopic images can assist in distinguishing visually similar lesion categories. This work evaluates MobileNetV2 and DINOv2 for classifying Basal Cell Carcinoma (BCC), Melanoma, and Nevus from a curated three-class subset of the ISIC 2019 dataset. The study focuses on a reproducible comparison of full-precision, quantized, and pruned models, together with a Grad-CAM explainability assessment. On the existing 2,489-image image-level validation partition, MobileNetV2 FP32 achieved 90.96% accuracy and 87.06% Melanoma recall. Full-integer INT8 TFLite conversion reduced the model size from 30.21 MiB to 2.94 MiB, while retaining 90.72% accuracy and 86.73% Melanoma recall; batch-one latency was 4.51 ms on the development CPU and 13.92 ms (71.8 FPS) on a Raspberry Pi 4 Model B, compared with 597.59 ms for the FP32 Keras model on the same device. In contrast, 50% unstructured pruning reduced accuracy to 83.49%, while native dynamic INT8 quantization of DINOv2 provided a compact 87.38 MiB representation and 173.54 ms host latency, with accuracy of 61.59% under the evaluated quantization configuration. This result identifies a clear opportunity for quantization-aware training and distillation. The two models are evaluated as independent branches; a post-hoc probability-fusion audit gave no reliable gain over MobileNetV2, so fusion is not part of the pipeline. All results are validation results, because lesion-level auditing identified 54.04% overlap between validation and training images. The results therefore support comparative compression and deployment analysis, but not clinical generalization. Grouped held-out test evaluation remains necessary before any clinical performance claim.

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Peer-review under responsibility of the scientific committee of the 5th International Conference on Machine Learning and Data Engineering (ICMLDE2026).

Keywords: Skin lesion classification; Model compression; MobileNetV2; DINOv2; Edge deployment; ISIC 2019

1. Introduction

BCC, Melanoma, and Nevus can look very similar under a dermatoscope, and a missed Melanoma is the most costly error a classifier can make. MobileNetV2 uses inverted residual blocks and depthwise separable convolutions to reduce computational complexity [1], whereas DINOv2 provides a transformer-based self-supervised representation [2]. Evaluating these complementary architectures is useful for understanding the accuracy–efficiency trade-off in resource-constrained applications.

We work with a curated three-class subset of ISIC 2019. The selected classes provide a focused evaluation of BCC, Melanoma, and Nevus, with particular attention to the clinically important confusion between Melanoma and Nevus. The experiments compare FP32 models, MobileNetV2 Dynamic-Range and Full INT8 TFLite models, a 50%

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Peer-review under responsibility of the scientific committee of the 5th International Conference on Machine Learning and Data Engineering (ICMLDE2026).

unstructured-pruned MobileNetV2, and DINOv2 dynamic INT8 diagnostics. The two architectures are evaluated as independent branches; a post-hoc probability-fusion audit is reported only to explain why fusion is excluded from the pipeline.

The main contributions of this work are summarized as follows:

- Development of a three-class skin lesion classification framework for BCC, Melanoma, and Nevus.
- Evaluation of MobileNetV2 and DINOv2 using 50-epoch training experiments.
- Analysis of training and validation accuracy and loss using the recorded epoch-wise experimental histories.
- Independent evaluation of the two branches, with a post-hoc audit showing that weighted probability fusion does not provide a reliable improvement over MobileNetV2.
- Quantitative comparison of model accuracy, class-level recall, storage size, and latency on a development CPU and a Raspberry Pi 4 across the evaluated model variants.
- Analysis of the trade-off between Full INT8 quantization and 50% unstructured pruning for MobileNetV2.
- Diagnosis of DINOv2 activation outliers that define the optimization target for quantization-aware training and distillation.
- Explicit auditing of lesion overlap and transparent separation of comparative validation evidence from independent clinical test evidence.
- Integration of the MobileNetV2 inference pipeline into a Streamlit-based application, and on-device benchmarking of the FP32 and quantized MobileNetV2 variants on a Raspberry Pi 4, including latency, memory, and sustained thermal behaviour.

2. Related Work

2.1. Convolutional Approaches to Skin Lesion Classification

Deep convolutional neural networks have demonstrated strong performance in automated skin lesion analysis. Esteva et al. demonstrated dermatologist-level skin cancer classification using deep neural networks, highlighting the potential of deep learning for dermatological image analysis [3].

Convolutional neural networks such as ResNet and MobileNet have subsequently been applied to image classification problems because of their ability to learn hierarchical visual features [4, 5]. MobileNetV2 improves computational efficiency using inverted residual structures and linear bottlenecks [1], making it suitable for applications requiring lightweight models.

2.2. Transformer-Based Visual Representation Learning

Vision Transformers provide an alternative approach to image analysis by modelling relationships between image regions using attention mechanisms [6]. Self-supervised learning approaches further improve the ability of models to learn transferable visual representations without requiring large quantities of labelled data.DINO introduced self-distillation for learning visual representations [7], while DINOv2 extended this approach to produce robust and transferable visual features suitable for multiple downstream computer vision tasks [2]. Such representations can be adapted for medical image classification through fine-tuning and task-specific classification layers.

2.3. Ensemble Learning

Ensemble learning combines predictions generated by multiple classifiers to improve overall performance. The potential benefit of an ensemble depends on the diversity of the individual models and the differences in their prediction errors [8, 9]. Combining convolutional and transformer-based architectures may provide complementary information because the two models learn visual representations using different architectural mechanisms. This work audits whether weighted probability-level fusion of MobileNetV2 and DINOv2 offers a reliable gain over the stronger single model.

3. Proposed Methodology

Figure 1 shows the full pipeline. The 12,445 ISIC 2019 images (BCC, Melanoma, and Nevus) are resized to 224×224, augmented, and normalized to the range 0–1, and are then divided by a stratified image-level split into 9,956 training and 2,489 validation images. Each preprocessed image then goes to two separate branches. The MobileNetV2 branch applies an ImageNet-pretrained [10] backbone, global average pooling, and dense layers of 256, 64, and 3 units. The DINOv2 branch converts the image into 256 patch embeddings and a CLS token, encodes them with 12 transformer blocks, and classifies the CLS representation with a linear head (768 → 3). For each branch, the figure also lists the FP32 results and the optimization experiments that followed. After Full INT8 TFLite quantization [11], MobileNetV2 still reached 90.72% accuracy, so we use it as the edge model; DINOv2 fell to 61.59% under INT8. We run Grad-CAM [12] on MobileNetV2 to see which lesion regions its predictions rely on.

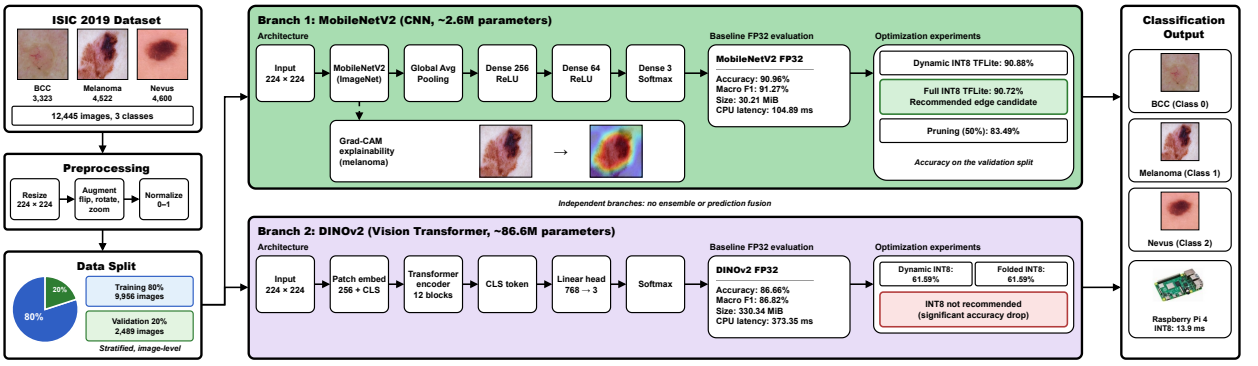


Fig. 1. DermaVision pipeline. Images are prepared and split, then classified by separate MobileNetV2 and DINOv2 branches, each shown with its FP32 results and optimization experiments. Grad-CAM explains the MobileNetV2 predictions, and the Raspberry Pi block shows the deployed Full INT8 model. Accuracies are on the 2,489-image validation split.

3.1. Dataset

We used a three-class subset of the ISIC 2019 skin lesion dataset [13], which combines images from the HAM10000 [14], BCN20000 [15], and ISIC 2017 challenge [16] collections. We kept three categories: Basal Cell Carcinoma (BCC), Melanoma, and Nevus. They cover a malignant melanocytic lesion, its most common benign look-alike, and the most common non-melanocytic skin cancer, which isolates the clinically most consequential Melanoma–Nevus confusion. With 3,323 BCC, 4,522 Melanoma, and 4,600 Nevus images, the largest class is only 1.38 times the smallest, so the subset is close to balanced. Both models used the same label order, so their probability outputs line up with the ground truth.

The class mapping used in this work is shown in Table 1.

Table 1. Class mapping used for skin lesion classification.

Class Label	Skin Lesion Category
0	BCC
1	Melanoma
2	Nevus

For final comparative evaluation, an image-level validation partition containing 2,489 images was used: 665 BCC, 904 Melanoma, and 920 Nevus images. The same class-label mapping and image ordering were maintained while generating predictions from MobileNetV2 and DINOv2 so that the predictions of both models could be compared image by image. This partition is not treated as an independent test set.

3.2. Data Splitting and Leakage Audit

The legacy evaluation protocol used a stratified image-level 80:20 split, producing 9,956 training images and 2,489 validation images. A subsequent lesion-ID audit identified 5,502 unique known lesions and 1,886 images without a

lesion ID. Of the validation images, 1,345 (54.04%) belonged to lesions also represented in the training partition. The overlap was particularly high for BCC (85.11% of validation images) and Melanoma (79.31%), while the corresponding Nevus overlap was 6.74%.

This overlap matters when reading the results. On the clean non-overlapping subset of 1,144 validation images, MobileNetV2 accuracy decreased from 90.96% to 86.80%, DINOv2 accuracy decreased from 86.66% to 84.00%, and the post-hoc equal-weight fusion decreased from 91.36% to 87.15%. We have prepared lesion-grouped splits for retraining. Until those models are trained, the numbers in this paper describe relative model and compression behaviour on the legacy validation partition.

3.3. Data Preprocessing

The input dermoscopic images were resized to 224×224 pixels. MobileNetV2 inputs were converted to RGB and scaled by $1/255$ to the range $[0, 1]$. For DINOv2 we normalized with the ImageNet mean and standard deviation. MobileNetV2 was trained with rotation, zoom, and translation augmentation. Horizontal flipping and brightness augmentation are not claimed for the reported run. The images were then converted to tensors and passed to each model.

3.4. MobileNetV2-Based Classification

The CNN branch is an ImageNet-pretrained MobileNetV2 [1] with a new three-class head for BCC, Melanoma, and Nevus. We trained it for 50 epochs and logged training and validation accuracy and loss after every epoch; these logs give the curves in Fig. 3. The recorded epoch history reached 90.12% validation accuracy at epoch 50; the saved FP32 checkpoint achieved 90.96% on the consolidated 2,489-image evaluation partition.

3.5. DINOv2-Based Classification

The transformer branch is a pretrained DINOv2 encoder with a three-class classification head, trained for 50 epochs with the same per-epoch logging. The recorded epoch history reached 86.82% validation accuracy at epoch 35; the saved FP32 checkpoint achieved 86.66% on the consolidated evaluation partition.

3.6. Post-hoc Fusion Audit

To test whether the two branches carry complementary information, their probability vectors were combined after training, as defined in Equation (1):

$$P_{\text{fused}} = w_M P_M + w_D P_D, \qquad w_M + w_D = 1, \qquad \hat{y} = \arg\max(P_{\text{fused}}) \tag{1}$$

Here, P_M and P_D are the probability vectors of MobileNetV2 and DINOv2, and w_M and w_D are their weights. This fusion is an audit only and is not part of the DermaVision pipeline shown in Figure 1. Its results are reported in Section 4.4.

3.7. System Implementation and Edge Deployment

We packaged the classifier as a Streamlit application. An uploaded dermoscopic image first passes an image quality check; accepted images are preprocessed and classified by the MobileNetV2 TFLite model. The interface then shows the predicted class together with the confidence score for each of BCC, Melanoma, and Nevus. Figure 2 illustrates the implemented application workflow, including the diagnostic interface, image quality assessment stage, and final prediction output.

The app currently runs the Dynamic-Range TFLite MobileNetV2 model. A separate Full INT8 TFLite inference module is available and is the recommended edge candidate; its Raspberry Pi 4 latency, memory, and thermal behaviour are reported in Section 4.6.

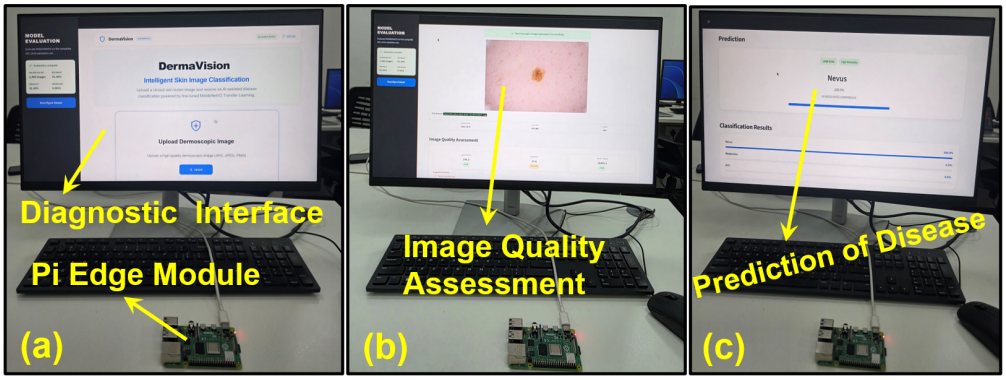


Fig. 2. Application workflow showing (a) the diagnostic interface and Pi Edge Module, (b) the image quality assessment stage, and (c) the final disease prediction with confidence scores.

4. Experimental Results and Analysis

4.1. Experimental Setup

We compared seven model variants on accuracy, precision, recall, F1-score, model size, and batch-one latency on the development CPU; the curves come from the per-epoch logs of the 50-epoch training runs. Section 4.6 gives the Raspberry Pi 4 measurements.

4.2. Individual Model Performance

Figure 3(a,b) shows how MobileNetV2 trained. Validation accuracy reached 90.12% at epoch 50, and the saved FP32 checkpoint scored 90.96% on the consolidated validation partition.

DINOv2 (Fig. 3(c,d)) peaked earlier, at 86.82% validation accuracy in epoch 35, and its saved FP32 checkpoint scored 86.66% on the same partition.

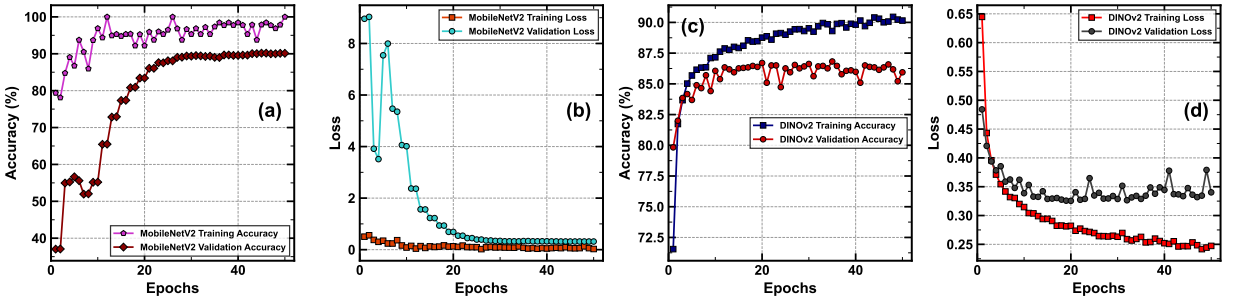


Fig. 3. Training and validation curves for 50 epochs. (a) MobileNetV2 accuracy; (b) MobileNetV2 loss; (c) DINOv2 accuracy; (d) DINOv2 loss. MobileNetV2 peaked at 90.16% validation accuracy in epoch 45 and ended at 90.12%; DINOv2 peaked at 86.82% in epoch 35.

4.3. Comparison of Individual Models

On the validation partition (Section 3.2), MobileNetV2 FP32 scored 90.96% accuracy, 91.27% Macro F1, and 87.06% Melanoma recall, against 86.66%, 86.82%, and 79.09% for DINOv2 FP32.

MobileNetV2 is more accurate and far smaller and faster, so it becomes the deployment branch; DINOv2 stays in the study as the transformer reference.

4.4. Post-hoc Fusion Audit Results

Table 2 sets MobileNetV2 against the two fusion weightings we tried. The previously reported configuration (w_M = 0.4, w_D = 0.6) achieved 90.76% accuracy and 85.18% Melanoma recall, both below MobileNetV2 alone. The equal-weight configuration (w_M = 0.5, w_D = 0.5) exceeded MobileNetV2 by only 0.40 percentage points, with weights evaluated on the same validation predictions and on a partition with documented lesion overlap. Because this gain is small, unverified on held-out data, and requires running the 330.34 MiB DINOv2 model alongside MobileNetV2, fusion is not adopted, and the remaining analysis focuses on the independent branches.

The two branches make partly different errors: DINOv2 correctly classified 110 images that MobileNetV2 misclassified, but misclassified 217 images that MobileNetV2 classified correctly. Equal-weight fusion corrected 27 MobileNetV2 errors and introduced 17 new ones, a net gain of 10 images, and raised recall in every class (BCC +0.30, Melanoma +0.33, Nevus +0.54 percentage points). An exact two-sided McNemar test on these discordant pairs gave p = 0.174, so the improvement is not statistically significant at the 0.05 level; the test also assumes independent samples, which the documented lesion overlap only partly satisfies. On the 1,144 non-overlapping validation images the gain was 0.35 percentage points (87.15% versus 86.80%).

Table 2. MobileNetV2 alone and two probability-fusion weightings on the validation partition.

Model / Configuration	Accuracy	Macro F1	Melanoma Recall
MobileNetV2 FP32 (adopted)	90.96%	91.27%	87.06%
Fusion audit (w_M = 0.5, w_D = 0.5)	91.36%	91.67%	87.39%
Fusion audit (w_M = 0.4, w_D = 0.6)	90.76%	91.01%	85.18%

4.5. Compression and Development-CPU Deployment Analysis

Table 3 lists the compression and deployment results. Full INT8 TFLite MobileNetV2 reduced storage by 90.27% relative to the FP32 checkpoint and retained accuracy within 0.24 percentage points. It also achieved 4.51 ms batch-one latency on the benchmark host. In contrast, 50% unstructured magnitude pruning [17] reduced accuracy by 7.47 percentage points and Melanoma recall by 13.61 percentage points without providing a dense CPU speedup. DINOv2 FP32 required 330.34 MiB on disk and showed 235.09–373.35 ms host latency. Native dynamic INT8 reduced DINOv2 storage to 87.38 MiB and latency to 173.54 ms, establishing a compact optimization baseline of 61.59% accuracy and 49.67% Melanoma recall for future quantization-aware improvement.

Table 3. Consolidated validation and development-CPU deployment measurements.

Model	Accuracy	Macro F1	Mel. Recall	Size / Latency
MobileNetV2 FP32	90.96%	91.27%	87.06%	30.21 MiB / 104.89 ms
MobileNetV2 Dynamic TFLite	90.88%	91.20%	85.62%	2.74 MiB / 27.76 ms
MobileNetV2 Full INT8 TFLite	**90.72%**	**91.03%**	**86.73%**	**2.94 MiB / 4.51 ms**
MobileNetV2 50% Pruned	83.49%	83.55%	73.45%	30.21 MiB / 101.54 ms
DINOv2 FP32	86.66%	86.82%	79.09%	330.34 MiB / 235–373 ms
DINOv2 Dynamic INT8	61.59%	57.54%	49.67%	87.38 MiB / 173.54 ms

The DINOv2 INT8 result provides a useful optimization diagnosis. Layer-wise diagnostics identified extreme activation outliers in transformer Blocks 8–11, a known obstacle to post-training transformer quantization [18, 19], with magnitudes up to approximately 494.72 compared with upper-percentile values near 1.3–2.7 in the corresponding diagnostic summaries. These observations identify the principal target for quantization-aware training, per-channel scaling, calibration refinement, or ViT-S distillation before deployment.

4.6. Raspberry Pi 4 On-Device Benchmark

All MobileNetV2 variants were benchmarked on a Raspberry Pi 4 Model B (Broadcom BCM2711, four Cortex-A72 cores at up to 1.8 GHz, 4 GB RAM, 64-bit Raspberry Pi OS, Python 3.13.5) without any hardware accelerator. TFLite models were executed with ai-edge-litert 2.3.0 (XNNPACK, four threads) and the FP32 Keras model with

TensorFlow 2.21.0 [20], using the same batch-one protocol as on the development CPU (50 warm-up and 200 timed runs on a preloaded tensor). The model files were byte-identical to those evaluated on the development CPU, so the accuracy values in Table 3 apply unchanged; every variant predicted the same class for the benchmark image. An unquantized FP32 TFLite conversion was also measured to separate the effect of the runtime from that of quantization. Table 4 summarizes the results.

Table 4. Raspberry Pi 4 Model B results for MobileNetV2 (batch one, four threads). Peak RSS is the memory of the whole Python process; Max temp. is the peak SoC temperature during a 300 s continuous run.

Variant	Disk	Latency (ms)	FPS	Peak RSS	Max temp.
FP32 Keras	30.21 MiB	597.59 ± 9.86	1.67	636.9 MiB	55.0 °C
FP32 TFLite (unquantized)	9.79 MiB	36.63 ± 0.34	27.30	69.5 MiB	74.5 °C
Dynamic-range INT8 TFLite	2.74 MiB	41.72 ± 3.42	23.97	59.2 MiB	72.0 °C
Full INT8 TFLite	**2.94 MiB**	**13.92 ± 0.20**	**71.83**	**51.7 MiB**	**78.4 °C**

Full INT8 TFLite ran at 13.92 ms per image (71.8 FPS), 42.9× faster than FP32 Keras and 2.6× faster than unquantized FP32 TFLite, while using the least memory. Most of the FP32 Keras latency is runtime overhead: converting the same FP32 weights to TFLite reduced latency from 597.59 ms to 36.63 ms. Dynamic-range quantization reduced storage to 2.74 MiB but was slightly slower than FP32 TFLite (41.72 ms), because its weights are dequantized during inference [21]; only full-integer quantization delivered a speedup on the Cortex-A72. With a single thread, Full INT8 required 45.86 ms and Dynamic-range 65.11 ms. Including decoding and preprocessing of a 1024×1024 PNG input, the Full INT8 end-to-end latency was 106.08 ms, so image decoding rather than inference dominates the on-device pipeline.

To check sustained behaviour, we ran each variant non-stop for 300 s, starting below 50 °C, and read the SoC temperature, ARM clock, and firmware throttling flags every 2 s. None of the runs triggered a throttling or under-voltage flag. Full INT8, which kept all four cores busy and completed 21,118 inferences, reached the highest temperature (78.4 °C), yet its mean latency was unchanged between the first and last 10% of the run (14.11 and 14.16 ms). FP32 Keras reached only 55.0 °C because it did not saturate the CPU.

4.7. Confusion Matrix and Error Analysis

Figure 4 presents the confusion matrix of MobileNetV2 FP32 for the three target classes: BCC, Melanoma, and Nevus. The diagonal elements represent correctly classified samples, while the off-diagonal elements represent misclassifications.

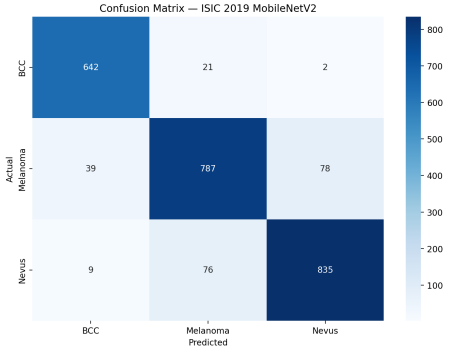


Fig. 4. Confusion matrix of MobileNetV2 FP32 for BCC, Melanoma, and Nevus classification on the 2,489-image validation partition.

For the BCC class, 642 images were correctly classified, 21 were predicted as Melanoma, and 2 were predicted as Nevus. For the Melanoma class, 787 images were correctly classified, 39 were predicted as BCC, and 78 were predicted as Nevus. For the Nevus class, 835 images were correctly classified, 9 were predicted as BCC, and 76 were predicted as Melanoma.

Table 5 presents the class-level precision, recall, and F1-score values derived from the confusion matrix.

Table 5. Class-level performance of MobileNetV2 FP32.

Class	Support	Precision	Recall	F1-score
BCC	665	93.04%	96.54%	94.76%
Melanoma	904	89.03%	87.06%	88.03%
Nevus	920	91.26%	90.76%	91.01%
Macro avg	2,489	91.11%	91.45%	91.27%

BCC achieved the highest recall, while Melanoma achieved the lowest. The confusion between Melanoma and Nevus is the main source of classification error: 78 Melanoma images were predicted as Nevus, and 76 Nevus images were predicted as Melanoma.

5. Discussion

The experimental results demonstrate that MobileNetV2 provides the strongest accuracy–efficiency trade-off among the evaluated variants. Its FP32 checkpoint achieved 90.96% validation accuracy and 87.06% Melanoma recall, while Full INT8 conversion retained 90.72% accuracy and 86.73% Melanoma recall with a substantially smaller model and lower host-CPU latency. On a Raspberry Pi 4 it ran at 71.8 FPS, which confirms Full INT8 MobileNetV2 as a practical edge model.

Fusion helped only a little. The 0.4/0.6 weighting (90.76%) stayed below MobileNetV2, and equal weighting added 0.40 percentage points, with weights chosen on the same validation predictions. The equal-weight fusion was also 7.79 percentage points less accurate on clean than on overlapping validation images. That gain does not justify running DINOv2 next to MobileNetV2 on an edge device, so the branches stay separate.

The class-level analysis shows that Melanoma remains the most important error category, particularly because Melanoma images are confused with Nevus. The 50% pruning experiment further shows that reducing parameter magnitudes without a suitable sparse execution strategy can harm this clinically relevant class. The DINOv2 INT8 experiment complements this result by identifying activation calibration as the main optimization bottleneck for transformer compression.

The framework is integrated into a Streamlit application, but the current application uses Dynamic-Range TFLite rather than the Full INT8 candidate. Apart from the McNemar test of the fusion audit, no confidence intervals, skin-tone fairness evaluation, or lesion-disjoint retrained test result are available in the present study. The dataset also contains only three of the eight ISIC 2019 categories, includes 1,886 images without lesion IDs, and has no external multi-centre validation. These limitations prevent clinical generalization.

6. Grad-CAM Explainability

Figure 5 shows Grad-CAM [12] heatmaps computed on the final convolutional layer of MobileNetV2 ($7\times7\times1280$) for one correctly classified image of each class and for a Melanoma image predicted as Nevus. In the correct cases, the activation is concentrated on the lesion rather than on the surrounding skin, hair, or imaging artefacts. In the failure case, the model also attends to the lesion but still predicts Nevus, which indicates that the error arises from how the lesion is interpreted rather than from attention to irrelevant regions. Grad-CAM is a coarse post-hoc attribution and does not establish clinical explainability.

7. Conclusion and Future Work

This work presented a comparative study of MobileNetV2 and DINOv2 for three-class skin lesion classification. On the legacy image-level validation partition, MobileNetV2 FP32 achieved 90.96% accuracy, while Full INT8 TFLite reduced storage to 2.94 MiB and retained 90.72% accuracy with 86.73% Melanoma recall. A post-hoc fusion audit showed no reliable gain over MobileNetV2 alone, so the two branches are kept independent. All results were obtained on a validation partition with 54.04% lesion overlap and therefore cannot be interpreted as independent test performance. The experiments established Full INT8 MobileNetV2 as the current practical edge baseline and identified the activation-calibration strategy required to improve DINOv2 compression.

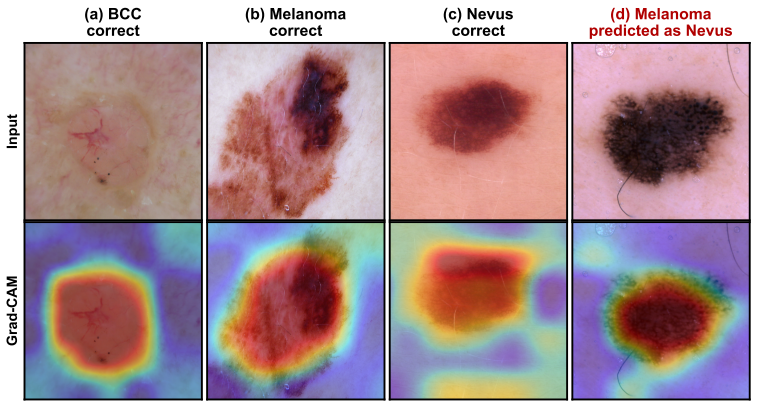


Fig. 5. Grad-CAM heatmaps of MobileNetV2 FP32: (a)–(c) correctly classified BCC, Melanoma, and Nevus images; (d) a Melanoma image predicted as Nevus. Top: input; bottom: heatmap overlay.

The main practical conclusion is that Full INT8 MobileNetV2 is the most promising candidate for immediate edge evaluation, while the DINOv2 INT8 result provides a measurable starting point for quantization-aware redesign and distillation. On a Raspberry Pi 4 it reached 13.92 ms per image (71.8 FPS) with 51.7 MiB peak memory, compared with 597.59 ms for FP32 Keras. The present results do not establish clinical generalization.

Future work will retrain both branches using the prepared zero-overlap lesion-grouped split, freeze any Melanoma safety threshold on validation data, re-test fusion on the lesion-disjoint split, and perform one final held-out test evaluation. Further work should include paired statistical tests, confidence intervals, power measurements and testing on additional edge devices, attention rollout for DINOv2, skin-tone fairness analysis, calibration, and comparison with external datasets. Extending the study to all eight ISIC 2019 categories, quantization-aware training, ViT-S distillation, out-of-distribution rejection, and imbalance-aware losses may improve the reliability of a future deployment-oriented system.

Acknowledgment

The authors would like to thank Amrita Vishwa Vidyapeetham, Chennai Campus, for providing the computing infrastructure and academic support for this work. The authors also acknowledge the AI DISC Lab and the NEGCES Lab for their support and facilities during the development and evaluation of this work.

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